

What we're doing

Collect distogram + structure (mmCIF) outputs for Protenix v2 on a subset of the FoldBench protein-monomer set, in single-sequence mode and with MSAs. Compute per-protein MAE (vs expected distances from the distogram) and dRMSD (CA-CA pairwise distances) and produce a comparison plot of single-seq vs MSA. Modal-hosted fan-out across `(protein × {single\_seq, msa})` on H100. Distograms captured via a forward hook on `runner.model.distogram\_head`.

Why

Eventual goal: compare MarFold's distograms head-to-head against Protenix in both modes. This experiment supplies the Protenix reference numbers under conditions MarFold will be scored on. No hypothesis on Protenix's numbers beyond "single-seq performs worse than MSA."

Reproducibility pins

FoldBench at commit ``4273f687...``. ``protenix==2.0.0``, ``torch==2.7.1``, ``gemmi==0.6.7``, Python 3.11.5. Protenix v2 weights from the HF mirror (``TMF001/pxdesign-weights``). MSAs pre-computed once via Protenix's ``colabfold`` backend (ColabFold MMseqs2 API), persisted to the weights Volume so reruns are deterministic and free. 5 seeds × 8 diffusion samples per (protein, mode); top-1 by Protenix's ``ranking_score``.

Results — distance metrics on CB-CB (in-range pair set)

GT in `[2.31, 21.84] Å` (Protenix v2's distogram range).

Mode	MAE CB	dRMSD CB	
-----	-----	-----	single_seq 2.69 / 2.96 3.75 / 4.15
msa	0.47 / 0.39	0.79 / 0.66	

(mean / median, in Å, lower is better)

Results — contact regime ($GT \leq 8 \text{ \AA}$)

Mode	MAE CB (contacts)	dRMSD CB (contacts)	
----- ----- -----			single_seq 3.56 / 3.97
5.49 / 6.18	msa	0.40 / 0.32	0.73 / 0.54

Results — structure & CASP contact precision

Median Kabsch CA-RMSD: single_seq ****14.4 Å****, msa ****1.18 Å**** — global-fold-correct in MSA mode, global-fold-wrong otherwise. CASP contact precision (long range) @ top L: msa ****0.99****, single_seq 0.28. LDDT-CA: msa 0.95, single_seq 0.42.

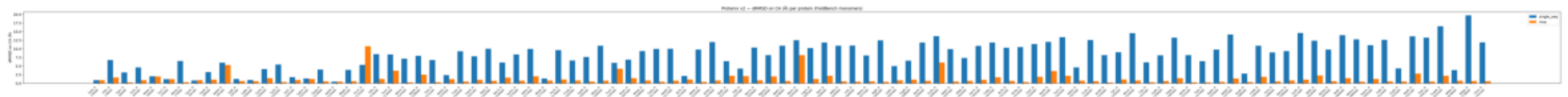
Conclusion

On 100 FoldBench monomers, Protenix v2 + MSA produces near-native structures (median CA-RMSD 1.18 Å, dRMSD 0.84 Å) — a regime where downstream MarinFold comparison is meaningful. Single-sequence mode degrades sharply on natural proteins (CA-RMSD 14.4 Å); only a viable MSA-free baseline for designed peptides. The distogram is a strong distance signal when the model is otherwise performing well; MSA-mode distogram MAE-CB is 0.47 / 0.39 Å (in-range filter).

For MarinFold cross-comparison, score against the same pair sets — intersection of Protenix's [2.31, 21.84 Å] with MarinFold's [0, 32 Å] is [2.31, 21.84 Å]. Raw outputs uploaded to HF at ``open-athena/MarinFold/data/protenix-foldbench-monomers/``.

drmsd_ca_angstrom_per_protein.png

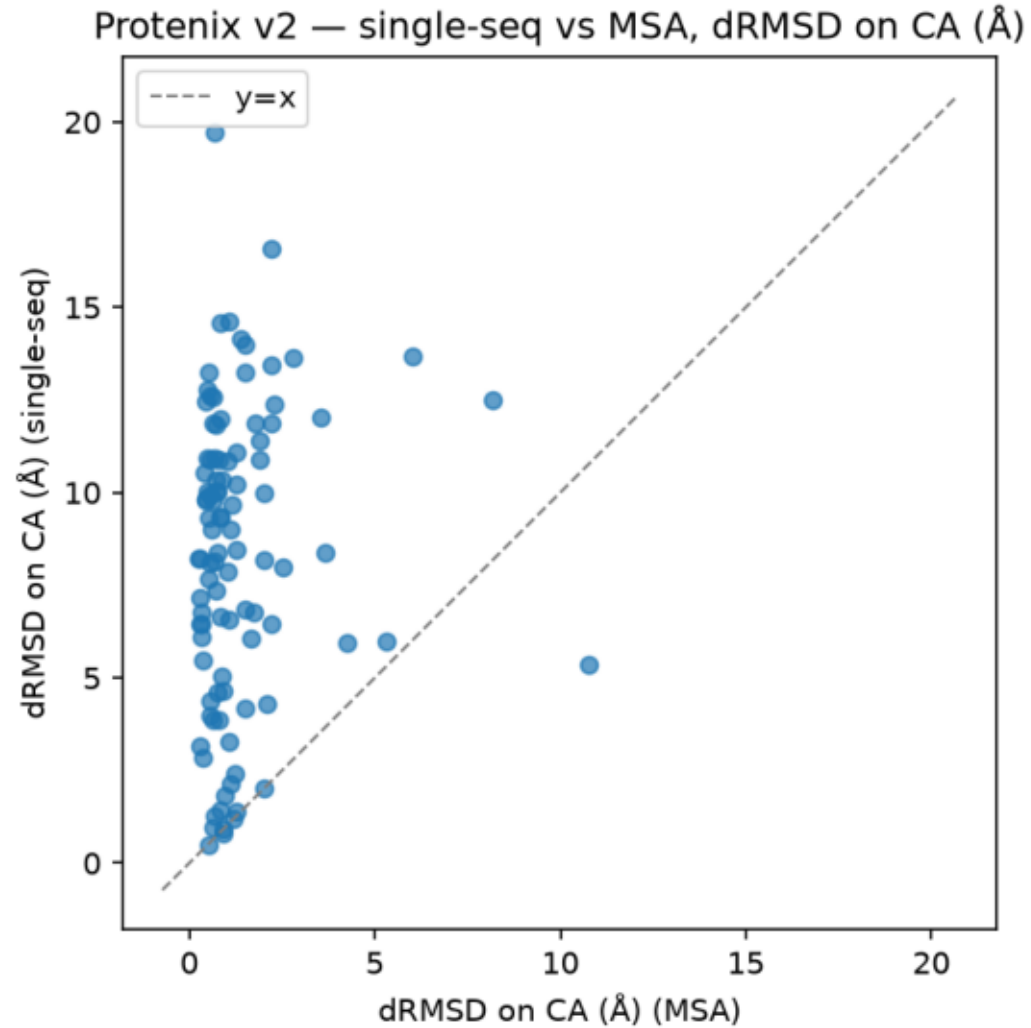
Per-protein dRMSD on CA (Å), single_seq vs MSA. Bars sorted by protein length. Top-1 sample per (protein, mode) by Protenix's ranking_score.



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$ cli.py plot --scores data/scores.csv --out plots/
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drmsd_ca_angstrom_ss_vs_msa_scatter.png

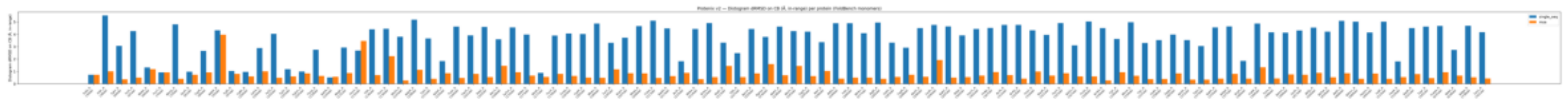
Paired dRMSD on CA (Å), $y=\text{single_seq}$, $x=\text{MSA}$. Each point is one protein; dashed line is $y=x$. Below the diagonal: MSA mode is better (typical for distance/structure metrics).



```
$ cli.py plot --scores data/scores.csv --out plots/
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drmsd_distogram_cb_angstrom_per_protein.png

Per-protein Distogram dRMSD on CB (Å, in-range), single_seq vs MSA. Bars sorted by protein length. Top-1 sample per (protein, mode) by Protenix's ranking_score.

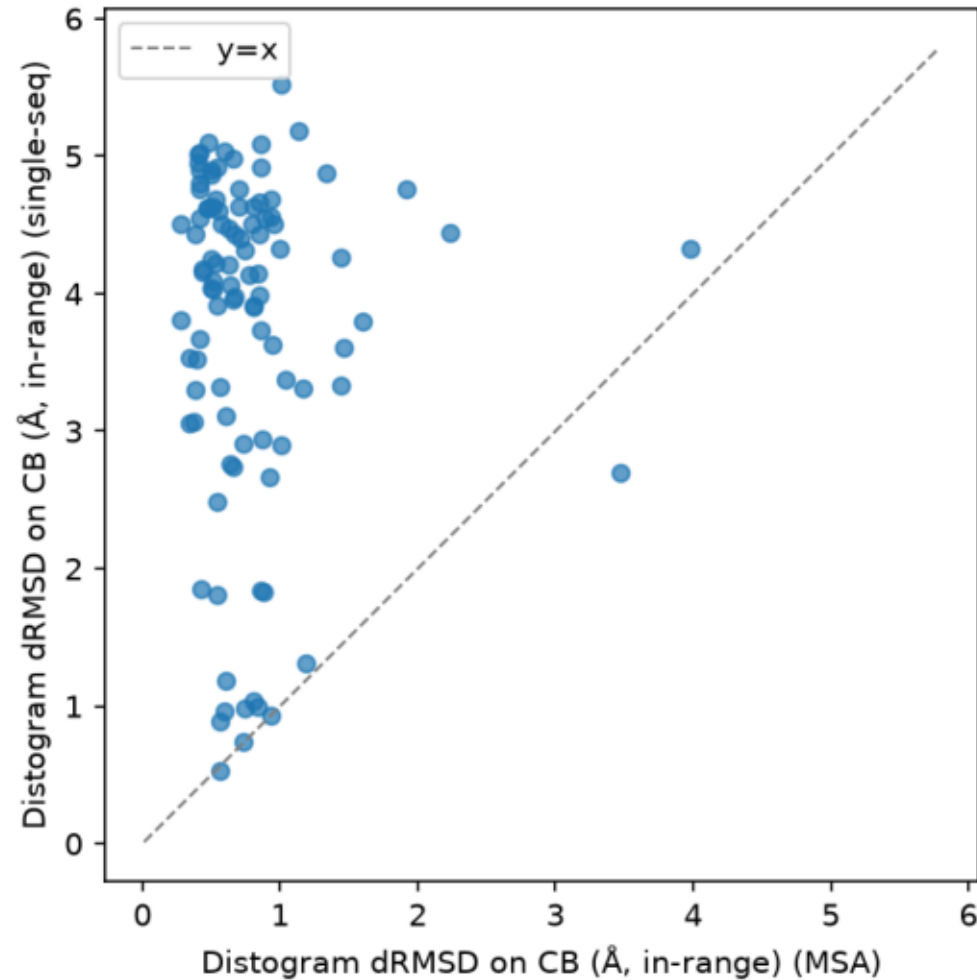


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$ cli.py plot --scores data/scores.csv --out plots/
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drmsd_distogram_cb_angstrom_ss_vs_msa_scatter.png

Paired Distogram dRMSD on CB (Å, in-range), y=single_seq, x=MSA. Each point is one protein; dashed line is $y=x$. Below the diagonal: MSA mode is better (typical for distance/structure metrics).

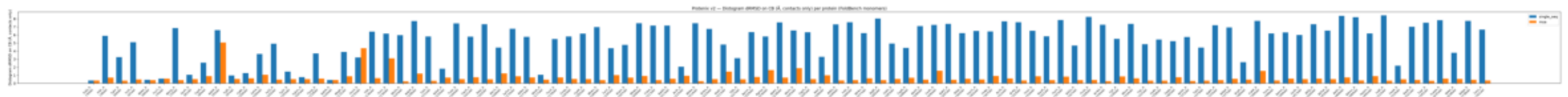
Protenix v2 — single-seq vs MSA, Distogram dRMSD on CB (Å, in-range)



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$ cli.py plot --scores data/scores.csv --out plots/
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drmsd_distogram_cb_contact_angstrom_per_protein.png

Per-protein Distogram dRMSD on CB (Å, contacts only), single_seq vs MSA. Bars sorted by protein length. Top-1 sample per (protein, mode) by Protenix's ranking_score.

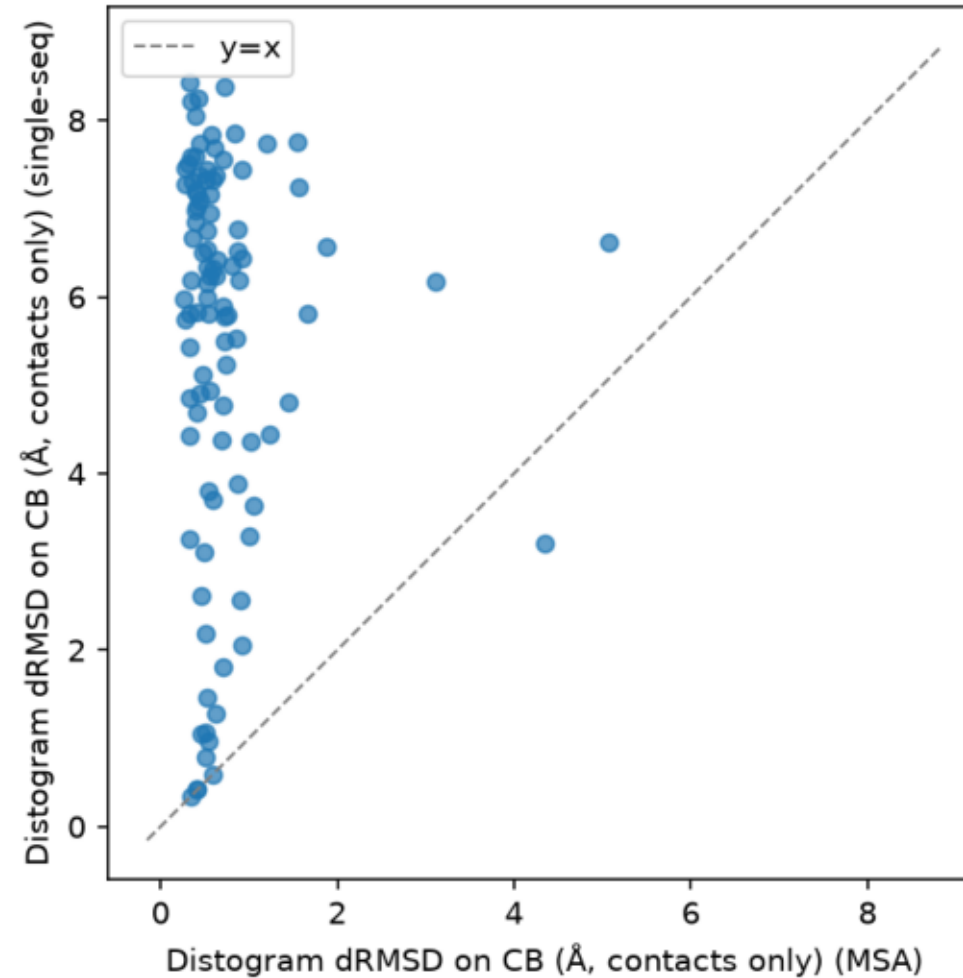


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$ cli.py plot --scores data/scores.csv --out plots/
```

drmsd_distogram_cb_contact_angstrom_ss_vs_msa_scatter.png

Paired Distogram dRMSD on CB ($\text{\AA}$, contacts only), $y=\text{single_seq}$, $x=\text{MSA}$. Each point is one protein; dashed line is $y=x$. Below the diagonal: MSA mode is better (typical for distance/structure metrics).

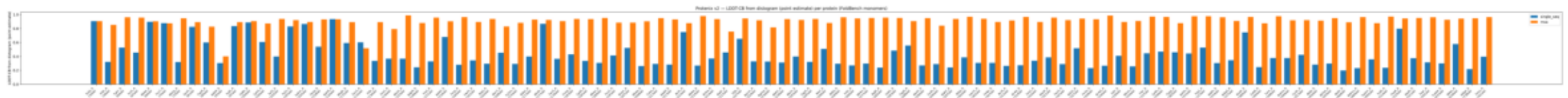
Protenix v2 — single-seq vs MSA, Distogram dRMSD on CB ($\text{\AA}$, contacts only)



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$ cli.py plot --scores data/scores.csv --out plots/
```

Iddt_distogram_cb_per_protein.png

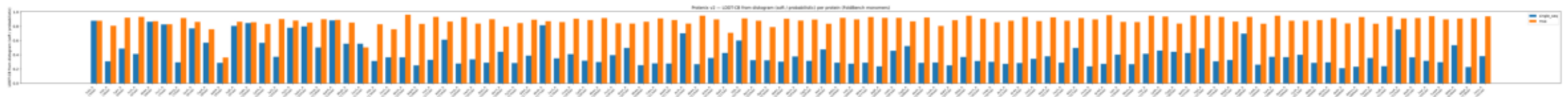
Per-protein LDDT-CB from distogram (point estimate), single_seq vs MSA. Bars sorted by protein length. Top-1 sample per (protein, mode) by Protenix's ranking_score.



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$ cli.py plot --scores data/scores.csv --out plots/
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Iddt_distogram_cb_soft_per_protein.png

Per-protein LDDT-CB from distogram (soft / probabilistic), single_seq vs MSA. Bars sorted by protein length. Top-1 sample per (protein, mode) by Protenix's ranking_score.

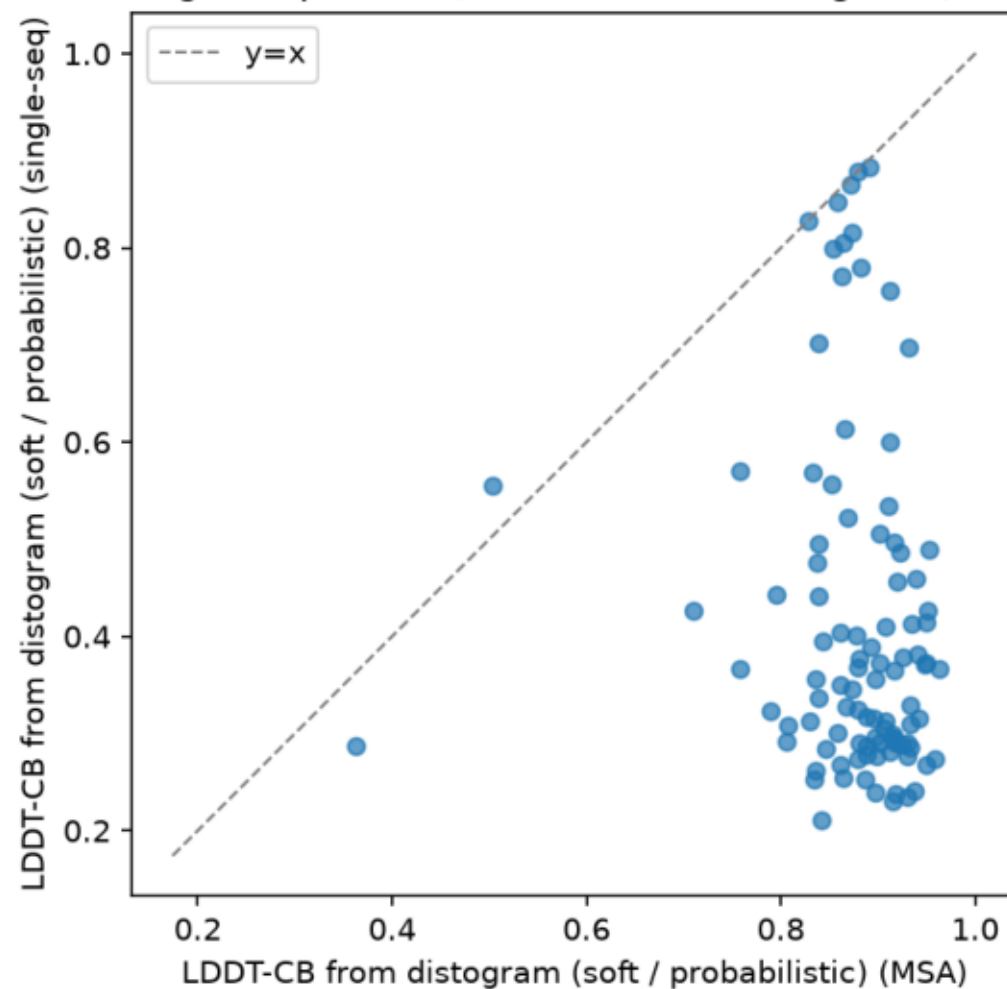


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$ cli.py plot --scores data/scores.csv --out plots/
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lddt_distogram_cb_soft_ss_vs_msa_scatter.png

Paired LDDT-CB from distogram (soft / probabilistic), y=single_seq, x=MSA. Each point is one protein; dashed line is $y=x$. Below the diagonal: MSA mode is better (typical for distance/structure metrics).

Protenix v2 — single-seq vs MSA, LDDT-CB from distogram (soft / probabilistic)

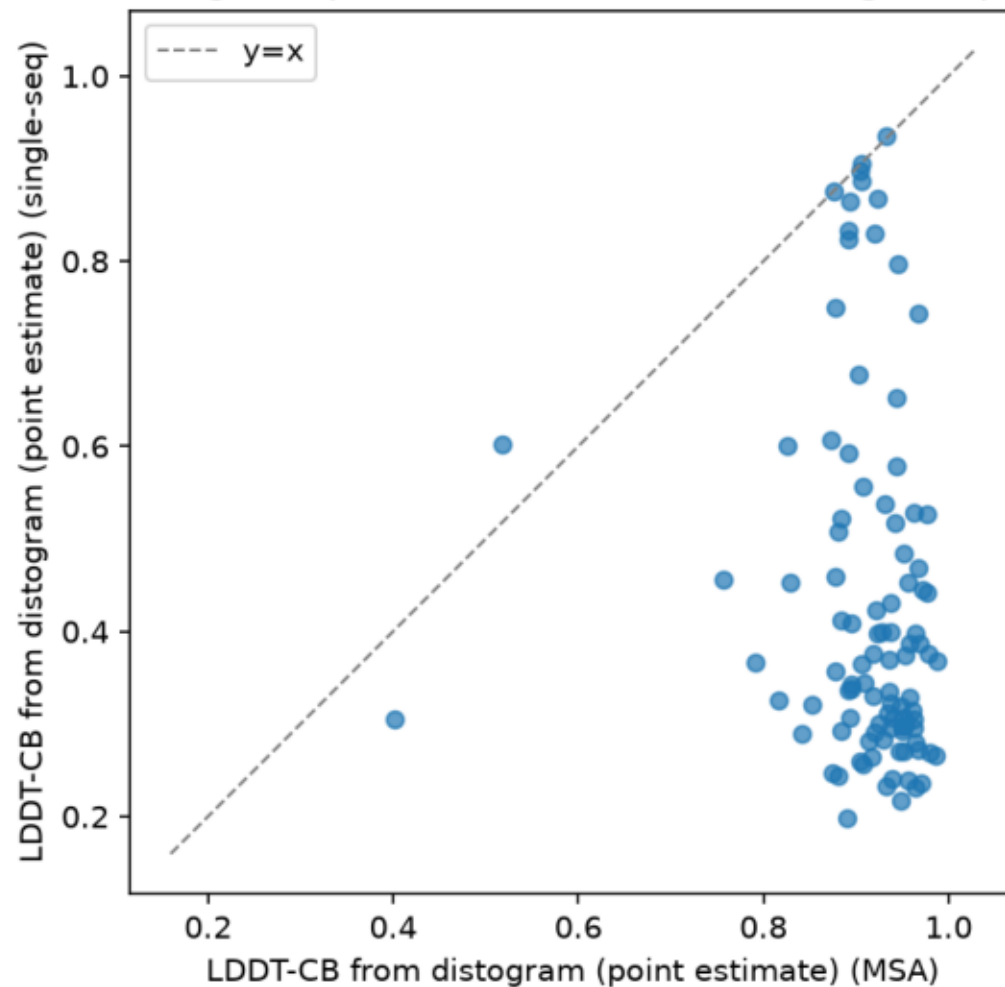


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$ cli.py plot --scores data/scores.csv --out plots/
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lddt_distogram_cb_ss_vs_msa_scatter.png

Paired LDDT-CB from distogram (point estimate), $y=\text{single_seq}$, $x=\text{MSA}$. Each point is one protein; dashed line is $y=x$. Below the diagonal: MSA mode is better (typical for distance/structure metrics).

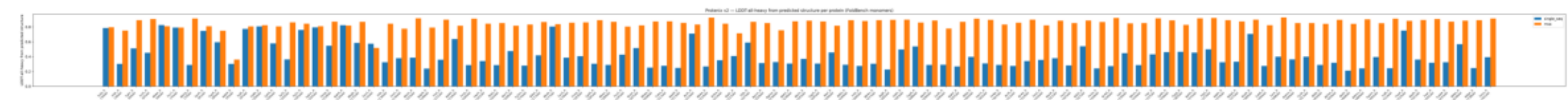
Protenix v2 — single-seq vs MSA, LDDT-CB from distogram (point estimate)



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$ cli.py plot --scores data/scores.csv --out plots/
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Iddt_structure_all_heavy_per_protein.png

Per-protein LDDT-all-heavy from predicted structure, single_seq vs MSA. Bars sorted by protein length. Top-1 sample per (protein, mode) by Protenix's ranking_score.

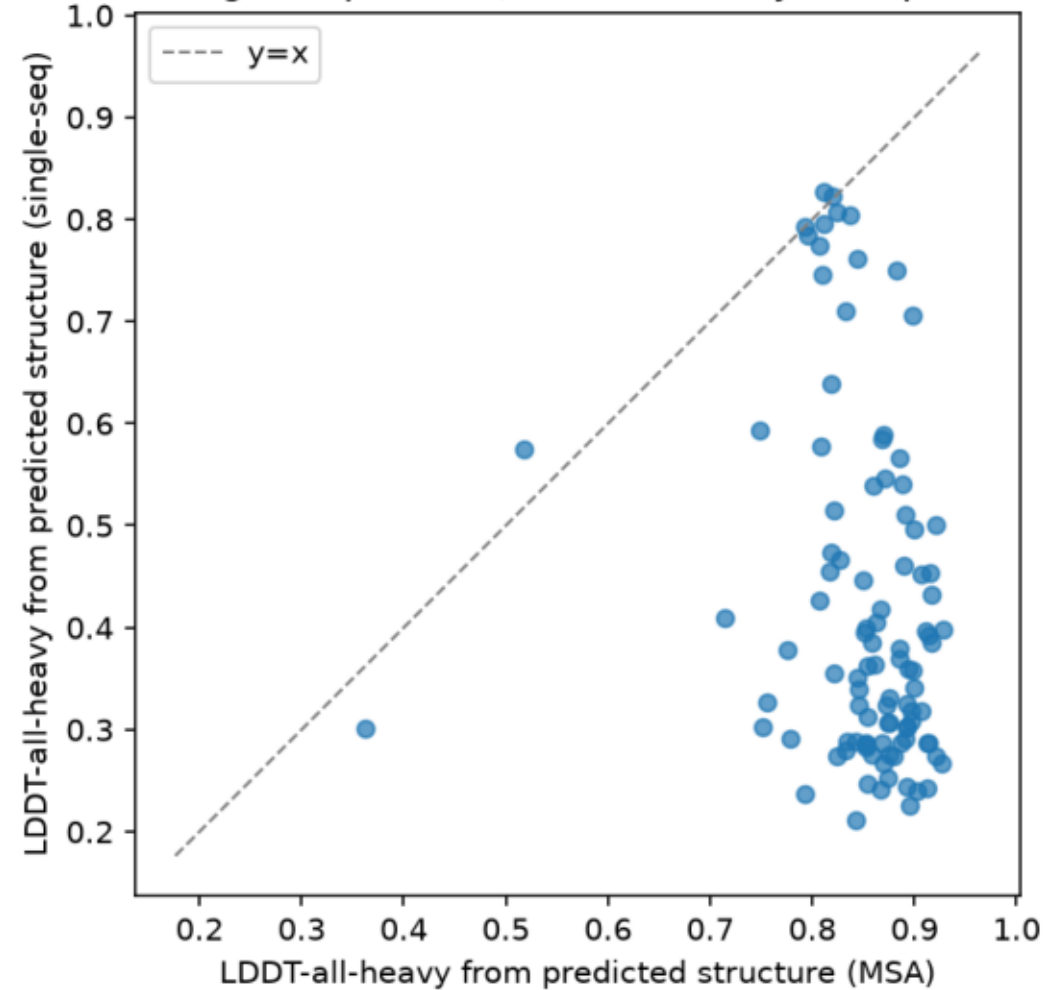


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$ cli.py plot --scores data/scores.csv --out plots/
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lddt_structure_all_heavy_ss_vs_msa_scatter.png

Paired LDDT-all-heavy from predicted structure, $y=\text{single_seq}$, $x=\text{MSA}$. Each point is one protein; dashed line is $y=x$. Below the diagonal: MSA mode is better (typical for distance/structure metrics).

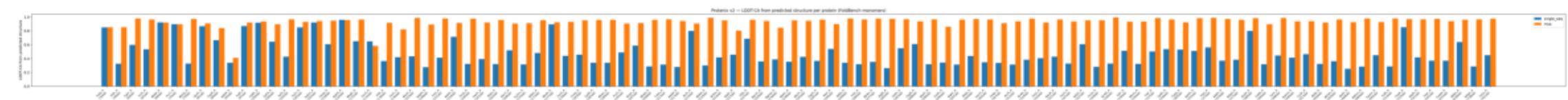
Protenix v2 — single-seq vs MSA, LDDT-all-heavy from predicted structure



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$ cli.py plot --scores data/scores.csv --out plots/
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Iddt_structure_ca_per_protein.png

Per-protein LDDT-CA from predicted structure, single_seq vs MSA. Bars sorted by protein length. Top-1 sample per (protein, mode) by Protenix's ranking_score.

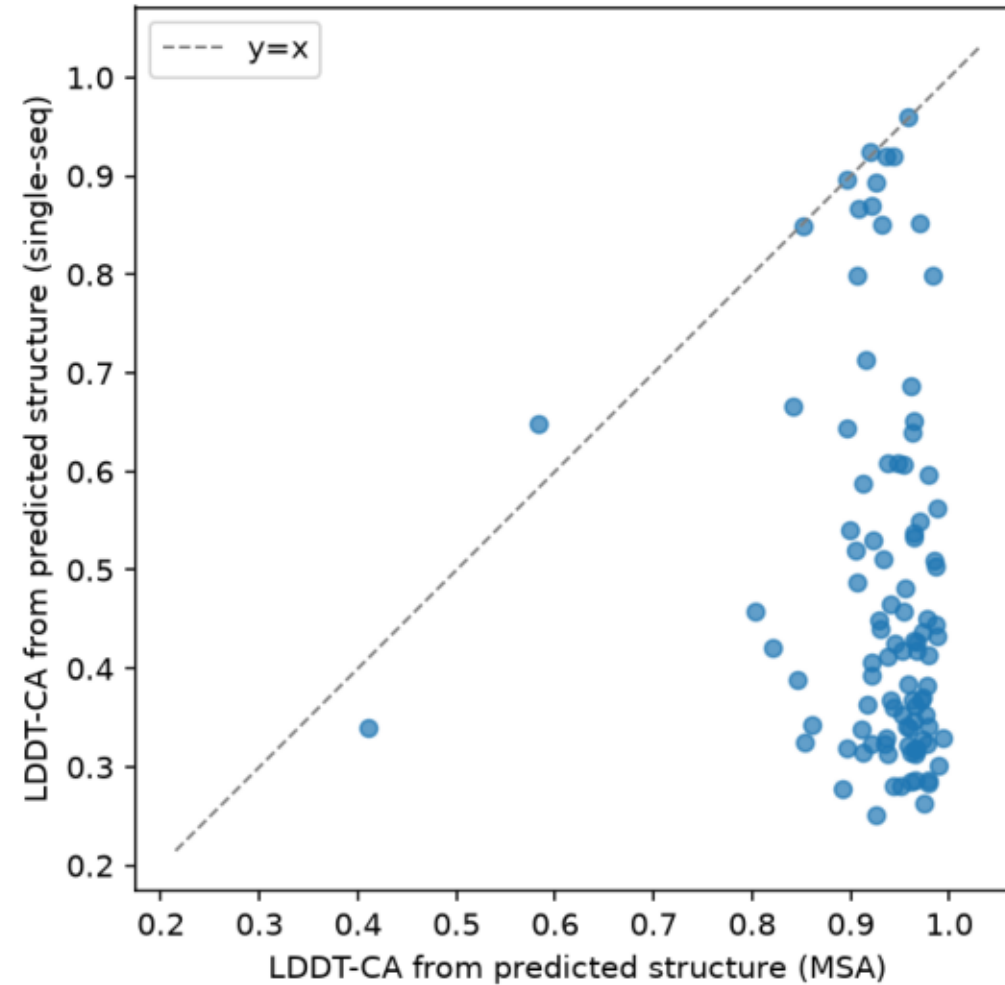


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$ cli.py plot --scores data/scores.csv --out plots/
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lddt_structure_ca_ss_vs_msa_scatter.png

Paired LDDT-CA from predicted structure, y=single_seq, x=MSA. Each point is one protein; dashed line is $y=x$. Below the diagonal: MSA mode is better (typical for distance/structure metrics).

Protenix v2 — single-seq vs MSA, LDDT-CA from predicted structure



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$ cli.py plot --scores data/scores.csv --out plots/
```

Iddt_structure_cb_per_protein.png

Per-protein LDDT-CB from predicted structure, single_seq vs MSA. Bars sorted by protein length. Top-1 sample per (protein, mode) by Protenix's ranking_score.

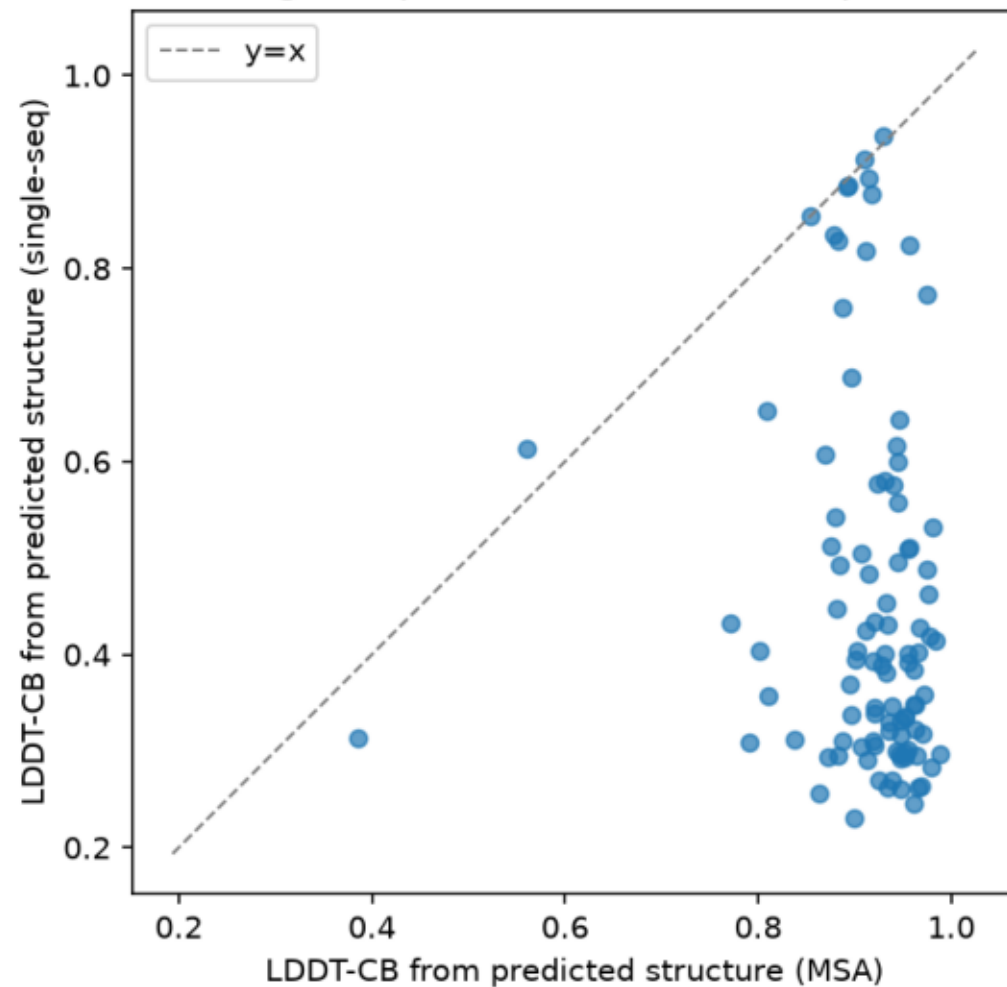


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$ cli.py plot --scores data/scores.csv --out plots/
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lddt_structure_cb_ss_vs_msa_scatter.png

Paired LDDT-CB from predicted structure, y=single_seq, x=MSA. Each point is one protein; dashed line is $y=x$. Below the diagonal: MSA mode is better (typical for distance/structure metrics).

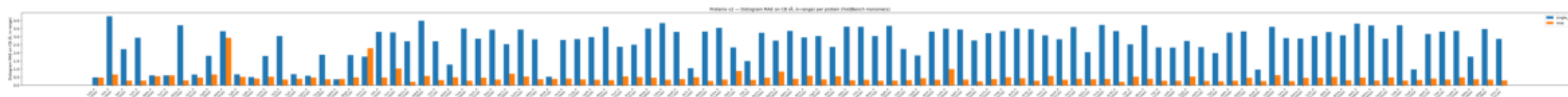
Protenix v2 — single-seq vs MSA, LDDT-CB from predicted structure



```
$ cli.py plot --scores data/scores.csv --out plots/
```

mae_distogram_cb_angstrom_per_protein.png

Per-protein Distogram MAE on CB (Å, in-range), single_seq vs MSA. Bars sorted by protein length. Top-1 sample per (protein, mode) by Protenix's ranking_score.

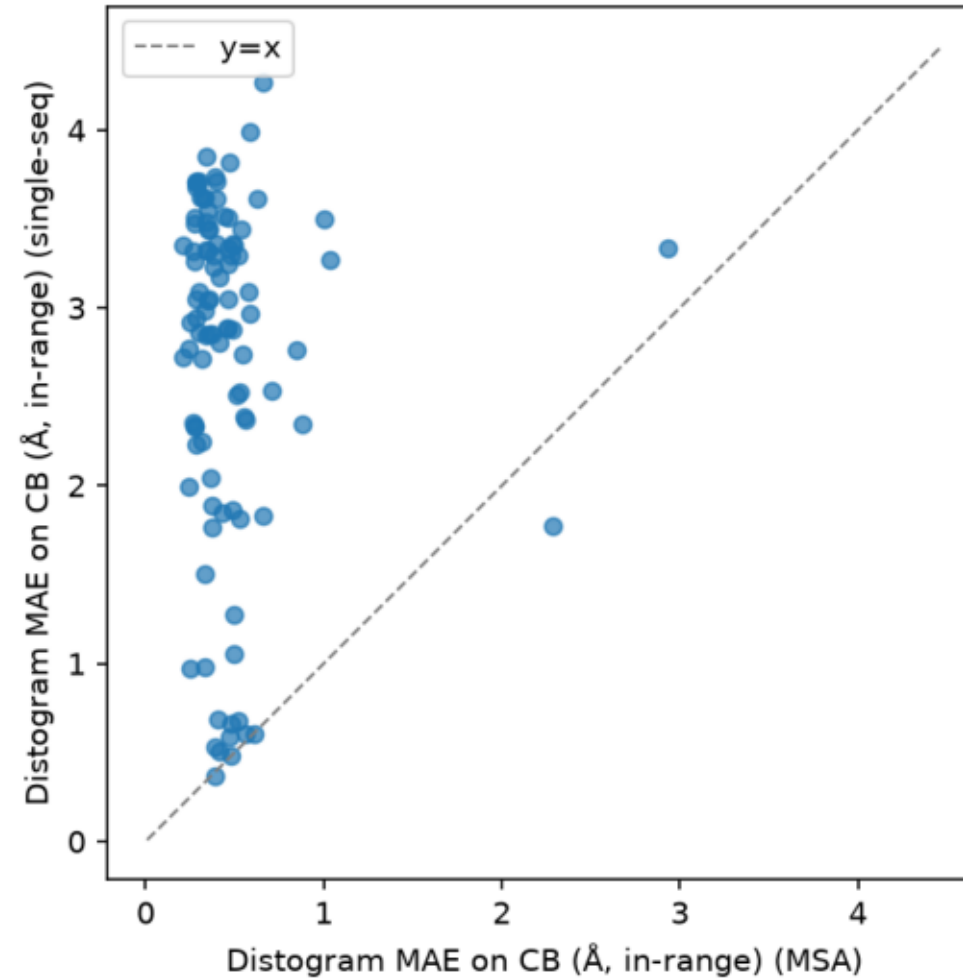


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$ cli.py plot --scores data/scores.csv --out plots/
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mae_distogram_cb_angstrom_ss_vs_msa_scatter.png

Paired Distogram MAE on CB (Å, in-range), y =single_seq, x =MSA. Each point is one protein; dashed line is $y=x$. Below the diagonal: MSA mode is better (typical for distance/structure metrics).

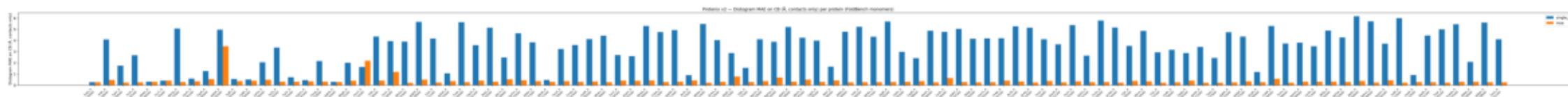
Proteinix v2 — single-seq vs MSA, Distogram MAE on CB (Å, in-range)



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$ cli.py plot --scores data/scores.csv --out plots/
```

mae_distogram_cb_contact_angstrom_per_protein.png

Per-protein Distogram MAE on CB ($\text{\AA}$, contacts only), single_seq vs MSA. Bars sorted by protein length. Top-1 sample per (protein, mode) by Protenix's ranking_score.

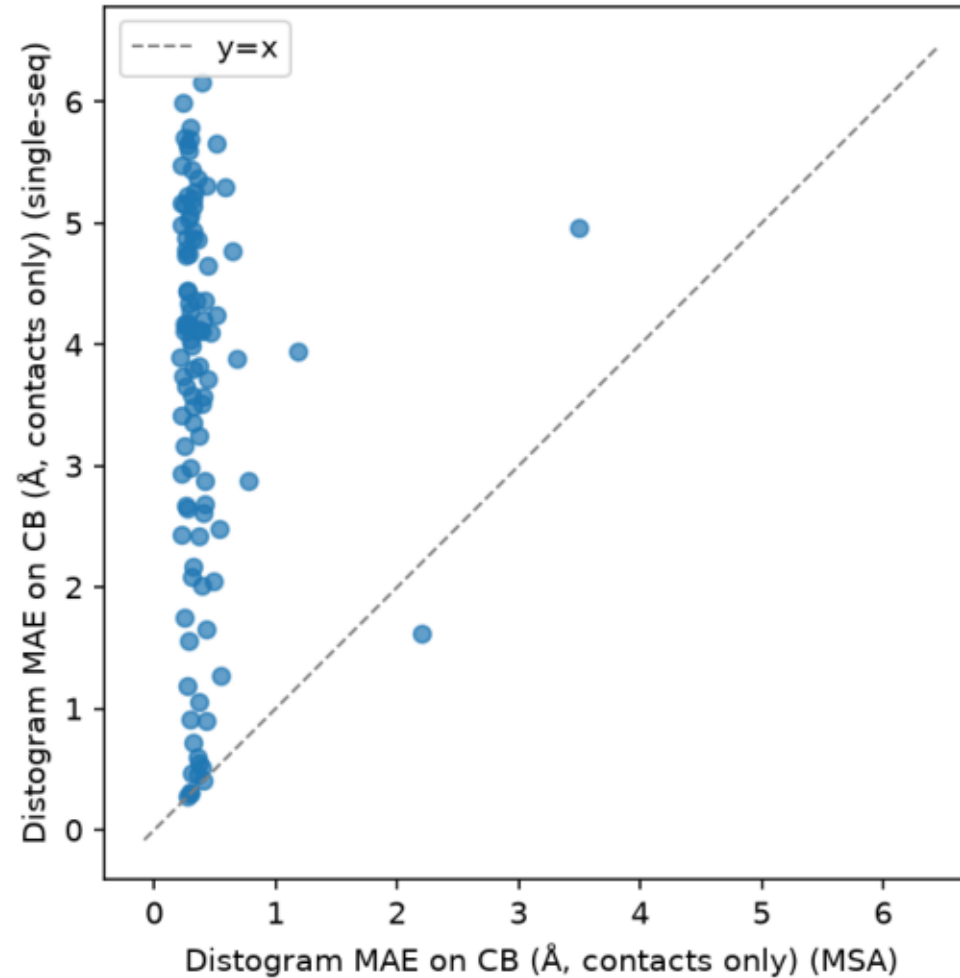


```
$ cli.py plot --scores data/scores.csv --out plots/
```

mae_distogram_cb_contact_angstrom_ss_vs_msa_scatter.png

Paired Distogram MAE on CB (Å, contacts only), y=single_seq, x=MSA. Each point is one protein; dashed line is $y=x$. Below the diagonal: MSA mode is better (typical for distance/structure metrics).

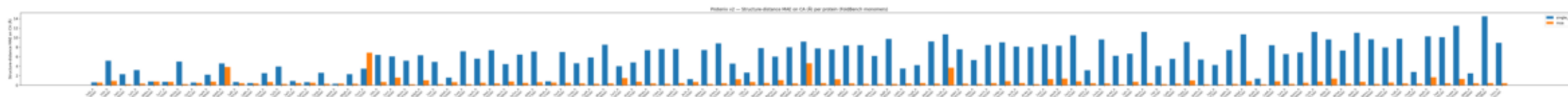
Protenix v2 — single-seq vs MSA, Distogram MAE on CB (Å, contacts only)



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$ cli.py plot --scores data/scores.csv --out plots/
```

mae_structure_ca_angstrom_per_protein.png

Per-protein Structure-distance MAE on CA (Å), single_seq vs MSA. Bars sorted by protein length. Top-1 sample per (protein, mode) by Protenix's ranking_score.

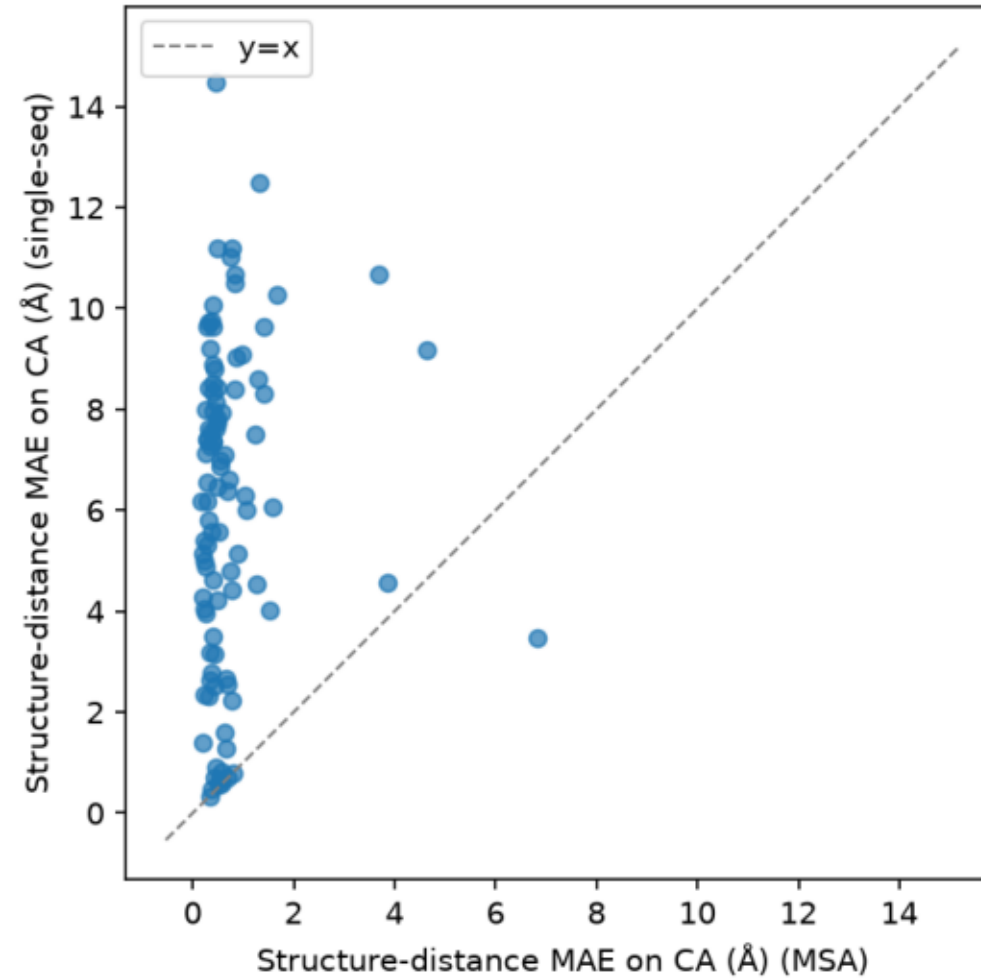


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$ cli.py plot --scores data/scores.csv --out plots/
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mae_structure_ca_angstrom_ss_vs_msa_scatter.png

Paired Structure-distance MAE on CA (Å), y =single_seq, x =MSA. Each point is one protein; dashed line is $y=x$. Below the diagonal: MSA mode is better (typical for distance/structure metrics).

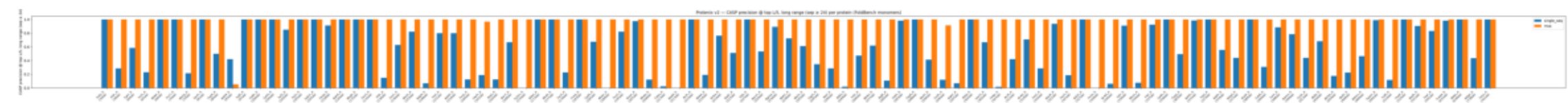
Protenix v2 — single-seq vs MSA, Structure-distance MAE on CA (Å)



```
$ cli.py plot --scores data/scores.csv --out plots/
```

prec_long_L_5_per_protein.png

Per-protein CASP precision @ top L/5, long range (sep ≥ 24), single_seq vs MSA. Bars sorted by protein length. Top-1 sample per (protein, mode) by Protenix's ranking_score.

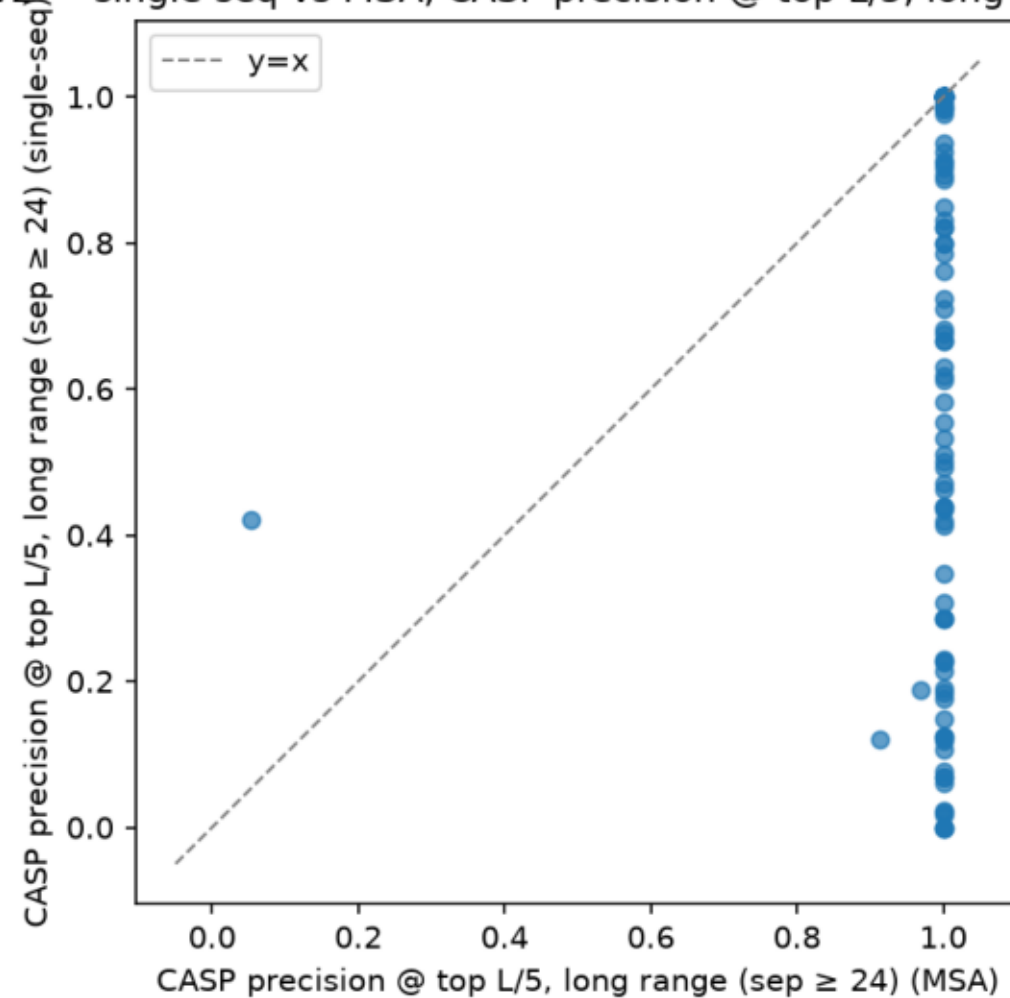


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$ cli.py plot --scores data/scores.csv --out plots/
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prec_long_L_5_ss_vs_msa_scatter.png

Paired CASP precision @ top L/5, long range (sep ≥ 24), y=single_seq, x=MSA. Each point is one protein; dashed line is y=x. Below the diagonal: MSA mode is better (typical for distance/structure metrics).

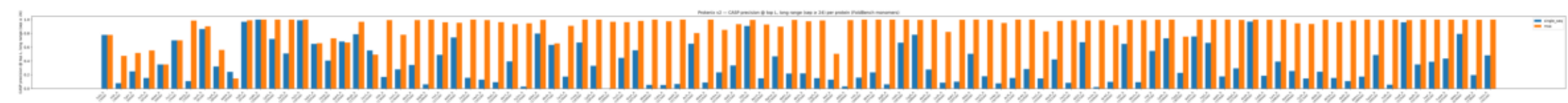
Protenix v2 — single-seq vs MSA, CASP precision @ top L/5, long range (sep ≥ 24)



```
$ cli.py plot --scores data/scores.csv --out plots/
```

prec_long_L_per_protein.png

Per-protein CASP precision @ top L, long range (sep ≥ 24), single_seq vs MSA. Bars sorted by protein length. Top-1 sample per (protein, mode) by Protenix's ranking_score.

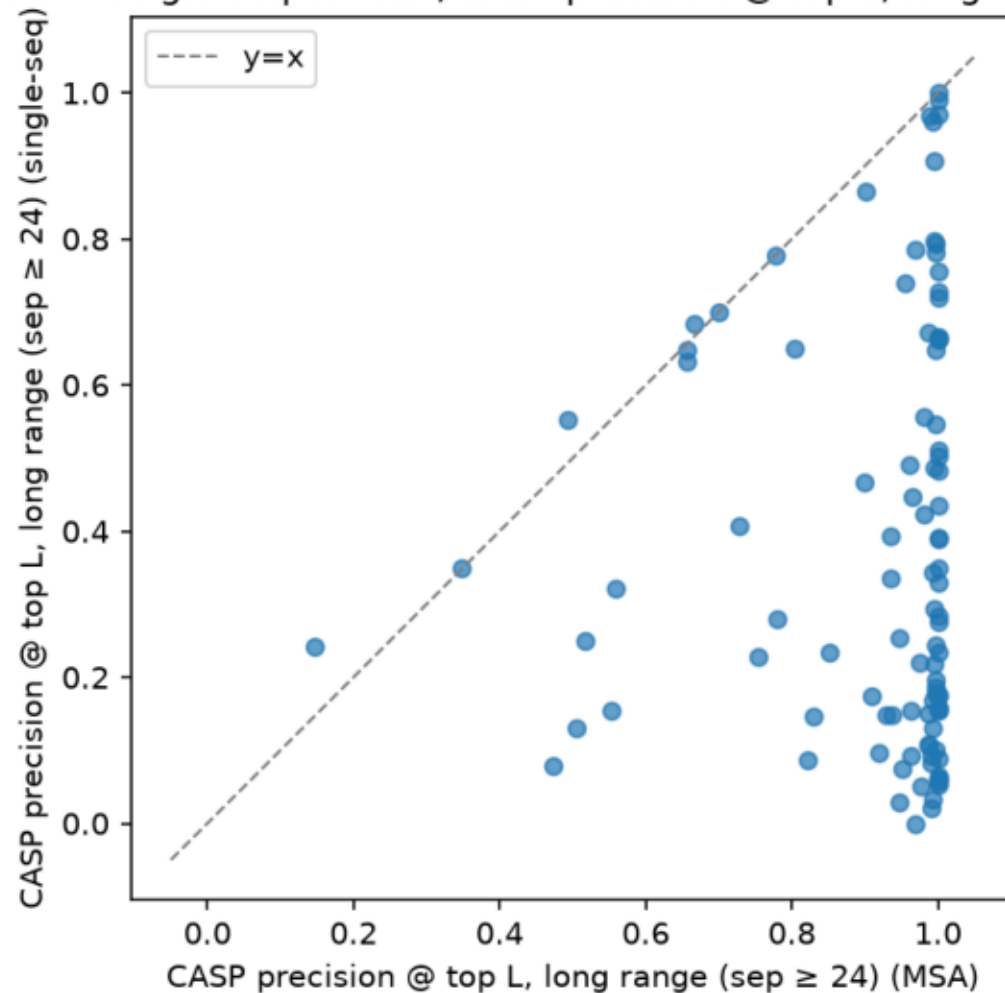


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$ cli.py plot --scores data/scores.csv --out plots/
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prec_long_L_ss_vs_msa_scatter.png

Paired CASP precision @ top L, long range (sep ≥ 24), y=single_seq, x=MSA. Each point is one protein; dashed line is y=x. Below the diagonal: MSA mode is better (typical for distance/structure metrics).

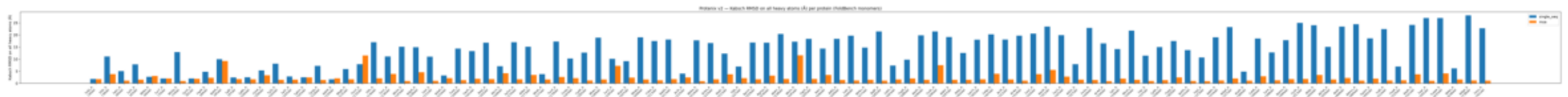
Protenix v2 — single-seq vs MSA, CASP precision @ top L, long range (sep ≥ 24)



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$ cli.py plot --scores data/scores.csv --out plots/
```

rmsd_all_heavy_angstrom_per_protein.png

Per-protein Kabsch RMSD on all heavy atoms (Å), single_seq vs MSA. Bars sorted by protein length. Top-1 sample per (protein, mode) by Protenix's ranking_score.

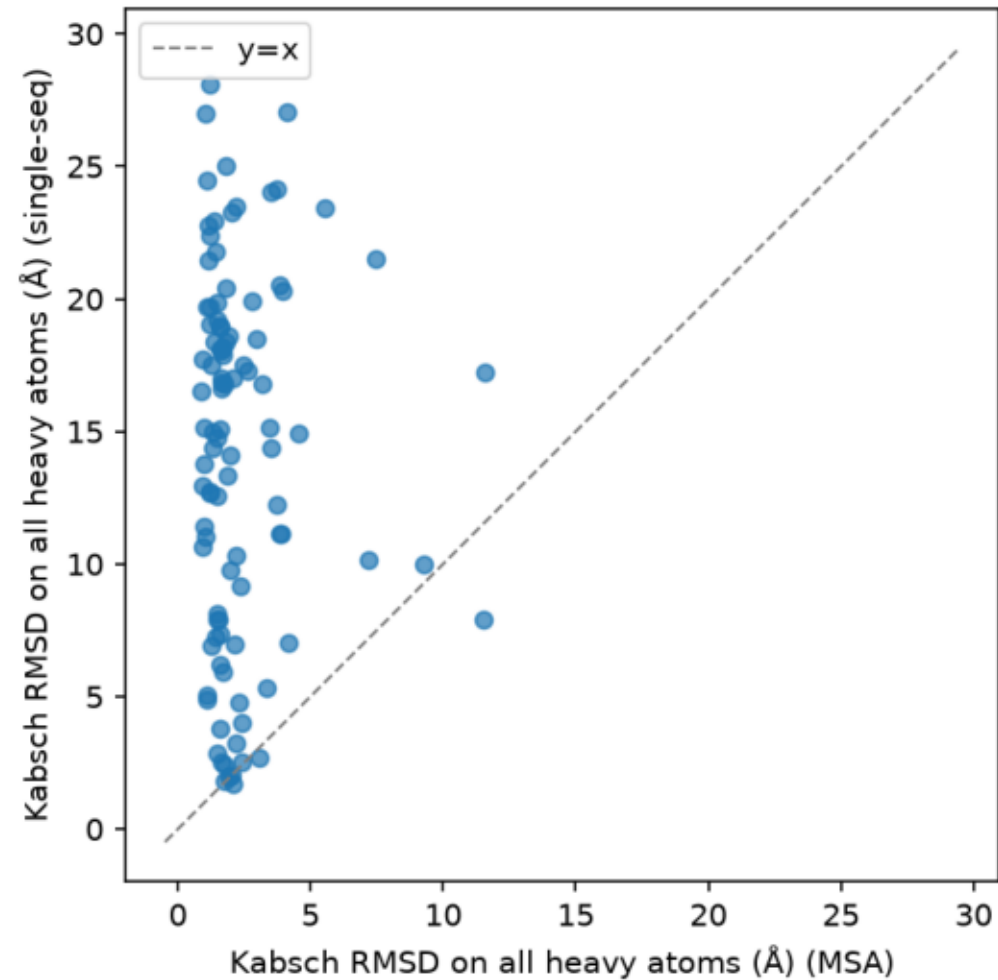


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$ cli.py plot --scores data/scores.csv --out plots/
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rmsd_all_heavy_angstrom_ss_vs_msa_scatter.png

Paired Kabsch RMSD on all heavy atoms (Å), y =single_seq, x =MSA. Each point is one protein; dashed line is $y=x$. Below the diagonal: MSA mode is better (typical for distance/structure metrics).

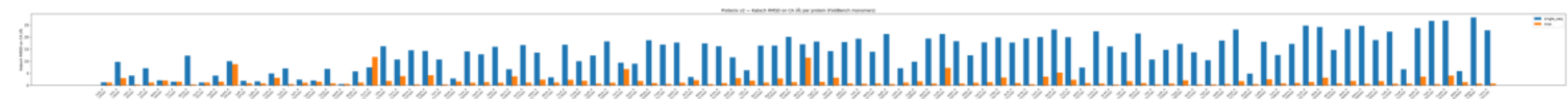
Protenix v2 — single-seq vs MSA, Kabsch RMSD on all heavy atoms (Å)



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$ cli.py plot --scores data/scores.csv --out plots/
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rmsd_ca_angstrom_per_protein.png

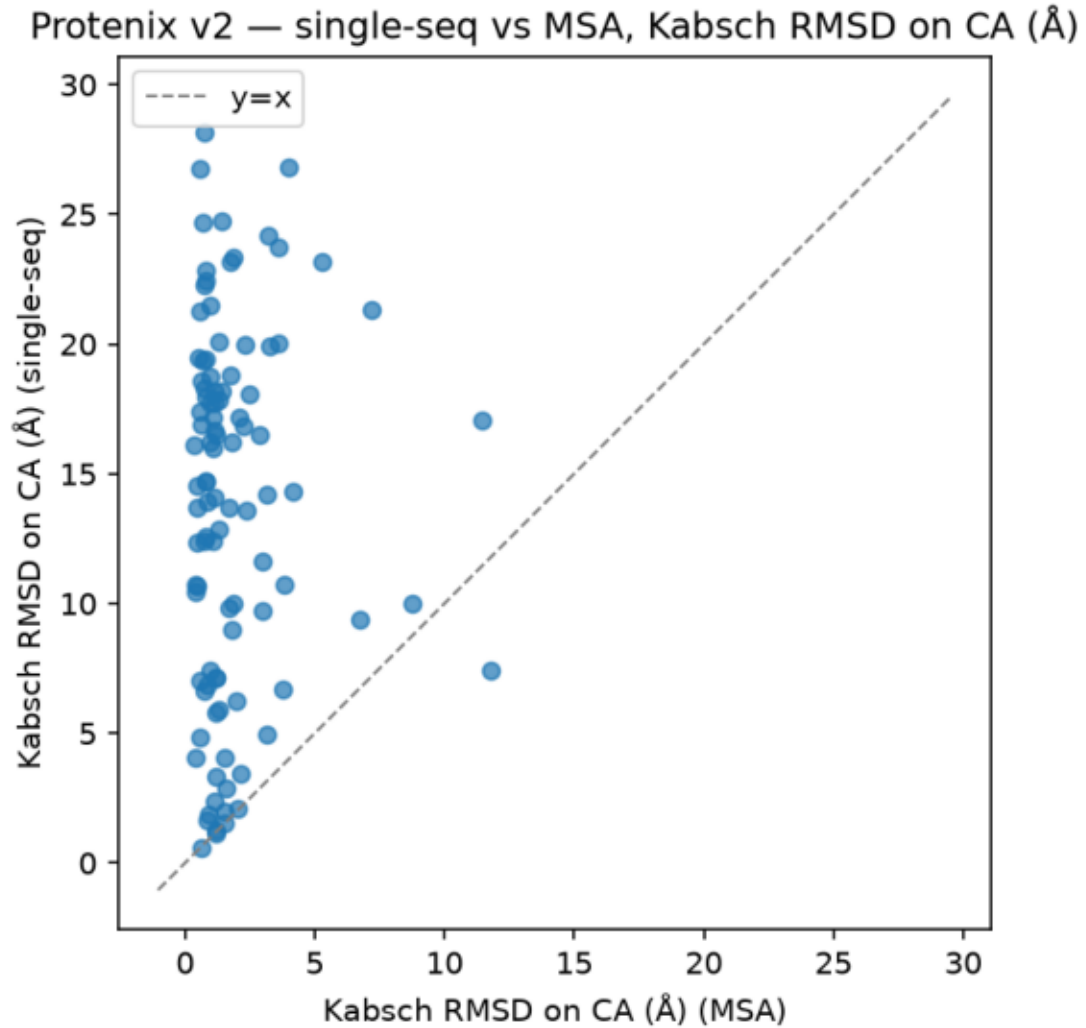
Per-protein Kabsch RMSD on CA (Å), single_seq vs MSA. Bars sorted by protein length. Top-1 sample per (protein, mode) by Protenix's ranking_score.



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rmsd_ca_angstrom_ss_vs_msa_scatter.png

Paired Kabsch RMSD on CA (Å), y =single_seq, x =MSA. Each point is one protein; dashed line is $y=x$. Below the diagonal: MSA mode is better (typical for distance/structure metrics).



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